DO NOW — RECALL

Magnetic, optical and flash storage

Name: _____ Class: _____ Date: _____

1. Why could a GPU be useful for many independent pixel operations?

2. What does ACC hold?

3. Define non-volatile.

4. Evaluate 11010110_2 AND 00001111_2, then write the resulting byte in hexadecimal.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It can execute many similar operations concurrently.

2. Model answer

Intermediate arithmetic/logical result.

3. Model answer

Data remains stored without power.

4. Model answer

$00000110_2 = 06_{16}$.

Vocabulary — know these terms

magnetic — Bits stored through magnetic states.

optical — Bits encoded/read through optical disc patterns.

flash — Non-volatile electronic storage.

seek latency — Delay in finding data.

backup — Separate recoverable copy of data.

NEW LEARNING

Read, annotate and explain

Identify the storage mechanism

HDDs use magnetised regions on rotating platters; DVD-style media are read optically.

Explain solid-state storage

Flash cells are non-volatile electronic storage without rotating platters or mechanical head seeks.

Evaluate rather than rank universally

SSD generally has low seek latency and resists mechanical shock; HDD may offer lower cost per GB at large capacities.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose SSD or HDD for a photographer moving large images between sites.

Predict

An SSD is attractive for frequent portable access and avoiding moving parts.

Explain

An HDD can offer capacity at a lower cost; backups remain necessary regardless.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give storage technology and one physical feature for HDD, DVD and USB flash.

Question 2

Explain two benefits and one drawback of an SSD rather than an HDD for frequent portable editing.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Propose fast working storage and bulk archival storage for a school, explaining backup strategy.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

USB flash and HDDs use the same magnetic storage method.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.